AgriTrace Project Overview & System Architecture

Full-Stack Decentralized Agri-Supply Chain Intelligence - Technical Presentation

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SLIDE 1: THE CORE PROBLEM

* Middlemen capture 50% to 65% of agricultural produce value.
* Consumers have zero reliable visibility into harvest origins, storage conditions, or chemical safety.
* Cold-chain failures cause perishable crop spoilage exceeding 30% during transit.

SLIDE 2: THE AGRITRACE SOLUTION

* Immutable on-chain provenance anchored to Ethereum Hardhat Local & Sepolia Testnet.
* IoT continuous cold-chain monitoring with automated mathematical excursion alerts.
* Role-based farmer and seller marketplace with authoritative database reward points.
* Multilingual accessibility with native Tamil, English, and Hindi interfaces.

SLIDE 3: PILOT RESULTS IN TAMIL NADU

* 200+ Agricultural datasets indexed across 10 crop categories in Tamil Nadu.
* Verified provenance validation with sub-second QR code lookups.
* Accessible Clear-Text and Voice AI navigation for rural smallholders.